

Quantum Mechanics III

Lecture 15

Blind-State Characterization

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$R_{final} = R_{QDC26} \cup P_1^*, \dots, P_4^*$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$F(\rho_i^*, \hat{\rho}_i) > F_{\text{registry}}$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$release = hash(QDCdiff, states, regression, log)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

Blind-State Characterization

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

Reading: Blind State Protocol; ICP-26 final checklist; Machine contribution profile